

RESEARCH OVERVIEW

I develop stable visual intelligence for dynamic worlds, connecting perception, understanding, and generation. My research studies how to preserve object identity, spatial relationships, and temporal consistency as observations and scenes change, with applications in autonomous systems, digital humans, and healthcare.

- AI for Digital Twins** 2025 - Present
  - World Models; Digital Humans
- AI for Autonomous Mobility** 2020 - Present
  - 3D Occupancy Prediction; Open-Vocabulary/Cross-Domain/Open-Set Object Detection
- AI for Scientific Innovation** 2024 - 2025
  - Nanophotonics; Meta-Optics; Graph-Based Learning
- AI for Transforming Medicine** 2020 - 2024
  - MRI Brain Analysis; Medical Report Generation; 4D Surgical Simulation

PROFILE HIGHLIGHTS

- Academic & Industrial Perspective**
  - Founded a seed-funded Hong Kong startup while pursuing top-tier research (50+ papers; 15+ lead-author papers).
- Large-Scale GPU Experimentation**
  - Extensive hands-on experience with large-scale GPU clusters (10,000+ GH200).
- Leadership & Collaboration**
  - Led a Swiss national AI initiative with 15 leading PIs; NeurIPS workshop organizer.
- Rapid Citation Impact**
  - U-KAN (co-first author) received 600+ citations within 1.5 years.
- Open-Source Impact**
  - Stable Video Infinity attracted **2.6k+** GitHub stars and was deployed on the Poe platform.

RESEARCH HIGHLIGHTS

- Oral/Spotlight/Highlight Presentations (First Author)**
  - AAAI 2022; CVPR 2022; ICCV 2023; ECCV 2024; ICCV 2025; NeurIPS 2025; ICLR 2026
- Best Paper Finalist (First Author)**
  - CVPR 2022 (33/8161)
- Full-Score Submission (First Author)**
  - ICCV 2025, final scores: 6, 6, 6
- Cover Paper (Second Author)**
  - ACS Photonics, 2024
- AAAI 2025 Top-1 Influential Paper (Co-First Author)**
  - U-KAN: medical image segmentation and generation backbone

WORK EXPERIENCE

- Ant Group** Jun. 16, 2026 - Present
  - Staff Research Engineer
  - Research areas: multimodal intelligence, world models, and AI for digital twins
- École Polytechnique Fédérale de Lausanne (EPFL)** Feb. 2025 - Jun. 2026
  - Research Scientist, Robotics and Computer Vision
  - Supervisor:** Prof. Alexandre Alahi (advised by Prof. Fei-Fei Li at Stanford)
- The Chinese University of Hong Kong (CUHK)** Mar. 2024 - Feb. 2025
  - Postdoctoral Researcher, Electronic Engineering
  - Supervisor:** Prof. Yixuan Yuan
- ScholaGO Education Technology Company Limited, Hong Kong** 2024 - Present
  - Founder & CEO
  - Developed multimodal educational products; secured seed funding from HK Tech 300 and HKSTP.

EDUCATION

- City University of Hong Kong (CityU)** Sep. 2020 - Feb. 2024
  - Ph.D., Electrical Engineering
  - 3.5 years; early graduation
  - Supervisor:** Prof. Yixuan Yuan
- Tianjin University (TJU)** Sep. 2016 - Jun. 2020
  - B.S., Electrical Engineering
  - GPA: 3.83/4.00 (Top 2%)

1. **[Generation] EverAnimate: Minute-Scale Human Animation via Latent Flow Restoration**  
**Wuyang Li**, Yang Gao, Mariam Hassan, Lan Feng, Wentao Pan, Po-Chien Luan, Alexandre Alahi  
 Conference on Neural Information Processing Systems (NeurIPS), 2026
2. **[Generation] SurgVista: Long-Horizon Surgical World Modeling with Plausible Instrument-Tissue Dynamics**  
 Wentao Pan\*, **Wuyang Li**<sup>†,\*</sup>, Shengyuan Liu, Xinyu Liu, Hengyu Liu, Yixuan Yuan  
 Conference on Neural Information Processing Systems (NeurIPS), 2026
3. **[Generation] Drift-Resistant Navigation World Model with Anchored Epipolar Guidance**  
 Po-Chien Luan, Zimin Xia, **Wuyang Li**, Yang Gao, Alexandre Alahi  
 Conference on Neural Information Processing Systems (NeurIPS), 2026
4. **[Generation] Proprio: Latent Self-Scoring and Inference-Time Refinement for Physically Plausible Video Generation**  
 Mariam Hassan, Kaouther Messaoud, **Wuyang Li**, Alexandre Alahi  
 Conference on Neural Information Processing Systems (NeurIPS), 2026
5. **[Generation] Anchored Video Generation: Decoupling Scene Construction and Temporal Synthesis in Text-to-Video Diffusion Models**  
 Mariam Hassan, Bastien Van Delft, **Wuyang Li**<sup>†</sup>, Alexandre Alahi  
 Under Review, 2026, High-scoring ECCV submission
6. **[Understanding] Social-Mamba: Socially-Aware Trajectory Forecasting with State-Space Models**  
 Po-Chien Luan, **Wuyang Li**<sup>†</sup>, Yang Gao, Alexandre Alahi  
 European Conference on Computer Vision (ECCV), 2026
7. **[Generation] Stable Video Infinity: Infinite-Length Video Generation with Error Recycling**  
**Wuyang Li**, Wentao Pan, Po-Chien Luan, Yang Gao, Alexandre Alahi  
 International Conference on Learning Representations (ICLR), 2026, **Oral (1.3%)**
8. **[Perception] Deformable Gaussian Occupancy: Decoupling Rigid and Nonrigid Motion with Factorized Distillation**  
 Yang Gao, **Wuyang Li**<sup>†</sup>, Po-Chien Luan, Alexandre Alahi  
 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2026
9. **[Perception] VoxDet: Rethinking 3D Semantic Occupancy Prediction as Dense Object Detection**  
**Wuyang Li**, Zhu Yu, Alexandre Alahi  
 Conference on Neural Information Processing Systems (NeurIPS), 2025, **Spotlight (3%)**
10. **[Perception] MetaScope: Optics-Driven Neural Network for Ultra-Micro Metalens Endoscopy**  
**Wuyang Li**<sup>\*</sup>, Wentao Pan\*, Xiaoyuan Liu\*, Zhendong Luo, Chenxin Li, Hengyu Liu, Din Ping Tsai, Mu Ku Chen, Yixuan Yuan  
 IEEE International Conference on Computer Vision (ICCV), 2025, **Final review scores (6, 6, 6)**
11. **[Perception] U-KAN Makes a Strong Backbone for Medical Image Segmentation and Generation**  
 Chenxin Li\*, Xinyu Liu\*, **Wuyang Li**<sup>\*</sup>, Cheng Wang\*, Hengyu Liu, Yifan Liu, Zhen Chen, Yixuan Yuan  
 AAAI Conference on Artificial Intelligence (AAAI), 2025, **AAAI 2025 Top-1 Influential Paper**
12. **[Perception] CLIFF: Continual Latent Diffusion for Open-Vocabulary Object Detection**  
**Wuyang Li**, Xinyu Liu, Jiayi Ma, Yixuan Yuan  
 European Conference on Computer Vision (ECCV), 2024, **Oral (1.8%)**
13. **[Generation] Boosting Medical Image Analysis via Motion-Informed Generative Videos**  
**Wuyang Li**, Xinyu Liu, Qiushi Yang, Yixuan Yuan  
 Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024
14. **[Perception] Novel Scenes & Classes: Towards Adaptive Open-set Object Detection**  
**Wuyang Li**, Xiaoqing Guo, Yixuan Yuan  
 IEEE International Conference on Computer Vision (ICCV), 2023, **Oral (1.8%)**
15. **[Perception] Adjustment and Alignment for Unbiased Open Set Domain Adaptation**  
**Wuyang Li**, Jie Liu, Bo Han, Yixuan Yuan  
 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2023
16. **[Perception] SIGMA++: Improved Semantic-complete Graph Matching for Domain Adaptive Object Detection**  
**Wuyang Li**, Xinyu Liu, Yixuan Yuan  
 IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2023

17. [\[Perception\]](#) **SIGMA: Semantic-complete Graph Matching for Domain Adaptive Object Detection**  
**Wuyang Li**, Xinyu Liu, Yixuan Yuan  
 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2022, **Best Paper Finalist (0.4%)**
18. [\[Perception\]](#) **SCAN: Cross Domain Object Detection with Semantic Conditioned Adaptation**  
**Wuyang Li**, Xinyu Liu, Xiwen Yao, Yixuan Yuan  
 AAAI Conference on Artificial Intelligence (AAAI), 2022, **Oral (4.5%)**
19. [\[Generation\]](#) **Synthesizing Realistic fMRI: A Physiological Dynamics-Driven Hierarchical Diffusion Model for Efficient fMRI Acquisition**  
 Yufan Hu, Yu Jiang, **Wuyang Li**, Yixuan Yuan  
 International Conference on Learning Representations (ICLR), 2025
20. [\[Generation\]](#) **InstantSplamp: Fast and Generalizable Stenography Framework for Generative Gaussian Splatting**  
 Chenxin Li\*, Hengyu Liu\*, Zhiwen Fan, **Wuyang Li**, Yifan Liu, Panwang Pan, Yixuan Yuan  
 International Conference on Learning Representations (ICLR), 2025
21. [\[Perception\]](#) **Universal Domain Adaptive Object Detection via Dual Probabilistic Alignment**  
 Yuanfan Zheng\*, Jinlin Wu\*, **Wuyang Li**, Zhen Chen  
 AAAI Conference on Artificial Intelligence (AAAI), 2025
22. [\[Perception\]](#) **Stereo Vision Meta-Lens-Assisted Driving Vision**  
 Xiaoyuan Liu, **Wuyang Li**, Takeshi Yamaguchi, Zihan Geng, Takuo Tanaka, Din Ping Tsai, Mu Ku Chen  
 ACS Photonics, 2024, **Selected as Cover Paper**
23. [\[Perception\]](#) **Flaws can be Applause: Unleashing Potential of Segmenting Ambiguous Objects in SAM**  
 Chenxin Li\*, Yuzhi Huang\*, **Wuyang Li**, Hengyu Liu, Xinyu Liu, Qing Xu, Zhen Chen, Yue Huang, Yixuan Yuan  
 Conference on Neural Information Processing Systems (NeurIPS), 2024
24. [\[Perception\]](#) **When 3D Partial Points Meets SAM: Tooth Point Cloud Segmentation with Sparse Labels**  
 Yifan Liu, **Wuyang Li**, Cheng Wang, Hui Chen, Yixuan Yuan  
 Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024, **Early Accept (14.0%)**
25. [\[Perception\]](#) **Latent Diffusion Label Rectification for Semi-supervised Medical Image Segmentation**  
 Xinyu Liu, **Wuyang Li**, Yixuan Yuan  
 Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024
26. [\[Understanding\]](#) **fMRI to T1w MRI Knowledge Transfer Network for Brain Multi-phenotype Prediction**  
 Zhibin He, **Wuyang Li**, Yu Jiang, Zhihao Peng, Pengyu Wang, Xiang Li, Tianming Liu, Junwei Han, et al.  
 Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024
27. [\[Perception\]](#) **LGS: A Light-weight 4D Gaussian Splatting for Efficient Surgical Scene Reconstruction**  
 Hengyu Liu\*, Yifan Liu\*, Chenxin Li\*, **Wuyang Li**, Yixuan Yuan  
 Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024
28. [\[Generation\]](#) **Endora: Video Generation Models as Endoscopy Simulators**  
 Chenxin Li, Hengyu Liu, Yifan Liu, Brandon Feng, **Wuyang Li**, Xinyu Liu, Zhen Chen, Jing Shao, Yixuan Yuan  
 Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024
29. [\[Understanding\]](#) **Foundation Model for Any Phenotype Prediction via fMRI to sMRI Knowledge Transfer**  
 Zhibin He, **Wuyang Li**, Yifan Liu, Xinyu Liu, Junwei Han, Tuo Zhang, Yixuan Yuan  
 IEEE Transactions on Medical Imaging (TMI), 2024
30. [\[Understanding\]](#) **MRM: Masked Relation Modeling for Medical Image Pre-Training with Genetics**  
 Qiushi Yang, **Wuyang Li**, Baopu Li, Yixuan Yuan  
 IEEE International Conference on Computer Vision (ICCV), 2023
31. [\[Perception\]](#) **Intervention & Interaction Federated Abnormality Detection with Noisy Clients**  
 Xinyu Liu, **Wuyang Li**, Yixuan Yuan  
 Medical Image Computing and Computer Assisted Intervention (MICCAI), 2022, **Oral, Early Accept**
32. [\[Perception\]](#) **Towards Robust Adaptive Object Detection under Noisy Annotations**  
 Xinyu Liu, **Wuyang Li**, Qiushi Yang, Baopu Li, Yixuan Yuan  
 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2022
33. [\[Perception\]](#) **HTD: Heterogeneous Task Decoupling for Two-Stage Object Detection**  
**Wuyang Li**, Zhen Chen, Baopu Li, Dingwen Zhang, Yixuan Yuan  
 IEEE Transactions on Image Processing (TIP), 2021

AWARDS

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|-----------------------------------------------------------------------------------|-------------|
| • <b>Outstanding Academic Performance Award, CityU</b>                            | 2022 - 2023 |
| • Outstanding research achievements                                               |             |
| • <b>Research Tuition Scholarship, CityU</b>                                      | 2022 - 2023 |
| • Outstanding research achievements                                               |             |
| • <b>First Place in the Polyp Detection Challenge</b>                             | 2021        |
| • 3rd International Endoscopy Computer Vision Challenge and Workshop (EndoCV2021) |             |
| • <b>Undergraduate National Scholarship</b>                                       | 2017 - 2018 |
| • The highest undergraduate honor for a student in China                          |             |
| • <b>Silver Award in Tianjin Advanced Mathematical Competition</b>                | 2017        |
| • Selected to represent Tianjin University                                        |             |

SERVICE

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|----------------------------------------------------------------------------------------------------------------|-------------|
| • <b>Swiss AI Initiative Project Lead [Link]</b>                                                               | 2025        |
| • Led a national project with 15 leading Swiss PIs; gained large-scale GPU cluster experience (10,000+ GH200). |             |
| • <b>NeurIPS 2024 Workshop Organizer [Link]</b>                                                                | 2024        |
| • Organized the Advancements in Multimodal Medical Foundation Models workshop                                  |             |
| • <b>Research Funding Proposals</b>                                                                            | 2022 - 2024 |
| • Assisted my advisor in preparing several competitive research funding proposals                              |             |
| • <b>Startup Funding [Link]</b>                                                                                | 2023        |
| • Secured seed funding from HK Tech 300 and HKSTP                                                              |             |
| • <b>Founder, ScholaGO Education Technology Company Limited, Hong Kong [Link]</b>                              | 2024        |
| • Founded a company developing multimodal educational products                                                 |             |
| • <b>Conference and Journal Reviewer</b>                                                                       |             |
| • ICLR/ICML/NeurIPS; CVPR/ICCV; TPAMI; IJCV; TIP; NeurIPS <b>Top Reviewer</b>                                  |             |